

A Literature Survey and Dataset Exploration on CIC-IDS2017 for Cybersecurity and Data Mining

1st Miray Pınarbaşı
Department of Computer
Engineering
Fenerbahçe University
Istanbul, Türkiye
miray.pinarbasi@stu.fbu.edu.tr

2nd Sıla Özgel
Department of Computer
Engineering
Fenerbahçe University
Istanbul, Türkiye
sila.ozgel@stu.fbu.edu.tr

3rd Furkan Kaya
Department of Computer
Engineering
Fenerbahçe University
Istanbul, Türkiye
furkan.kaya@stu.fbu.edu.tr

4th Umut Topatan
Department of Computer
Engineering
Fenerbahçe University
Istanbul, Türkiye
umut.topatan@stu.fbu.edu.tr

5th Göktuğ Karaca
Department of Computer
Engineering
Fenerbahçe University
Istanbul, Türkiye
goktug.karaca@stu.fbu.edu.tr

6th Ezgi Erdoğan
Department of Computer
Engineering
Fenerbahçe University
Istanbul, Türkiye
ezgi.erdogan1@stu.fbu.edu.tr

Abstract—In this report we have provided includes a Literature Survey and an overall look at the Dataset CIC-IDS2017 and its uses, why it is being used and it's features. Namely how the dataset has been used and benefited from in the past and how it is used in current studies. We look at what the dataset consists of and its relevance to cybersecurity.

Index Terms—Machine Learning, Data Mining, Cyber Security, Intrusion detection.

I. OVERVIEW OF THE DATASET

A. Source/Purpose and Relevance to Cybersecurity

The CIC-IDS2017 dataset was developed by the Canadian Institute for Cybersecurity (CIC) at the University of New Brunswick, Canada, under the supervision of Ali A. Ghorbani, with principal contributors Iman Sharafaldin and Arash Habibi Lashkari. [2] It was published in 2018 after the ICISSP2018 Conference in Portugal. [39] The dataset we have chosen is one of the most widely used attack detection datasets in modern studies. As stated in the name it was developed in 2017. The dataset was captured during July 3–7, 2017, over a real corporate network environment consisting of 25 users, two independent LANs (attacker and victim networks), and typical enterprise components such as firewalls, routers, and switches.

Using the B-Profile System (Sharafaldin et al., 2016), the researchers simulated realistic human interactions to generate authentic benign background traffic, while launching several modern cyberattacks including Brute Force (SSH/FTP), Heartbleed, DoS/DDoS, Web attacks (SQLi, XSS), Infiltration, and Botnet ARES.

The project aimed to create a comprehensive, labeled, and realistic intrusion detection benchmark, overcoming the major limitations of older datasets such as DARPA-98 or

KDD'99, which lacked diversity and modern attack coverage.

Intrusion Detection Systems (IDSs) and Intrusion Prevention Systems (IPSs) are the most important defense tools against sophisticated and ever-growing network attacks. Due to the lack of reliable test and validation datasets, anomaly-based intrusion detection approaches are suffering from consistent and accurate performance evolutions.

The evaluations of the eleven existing datasets since 1998 show that most are out of date and unreliable. Some of these datasets suffer from the lack of traffic diversity and volumes, some do not cover the variety of known attacks, while others anonymize packet payload data, which cannot reflect the current trends. Some are also lacking feature set and metadata. The total dataset is approximately 50 GB.

B. Features in the Dataset

CIC-IDS2017 contains both raw packet capture (PCAP) files and pre-processed CSV data representing labeled flow-based features extracted using CICFlowMeter-V3. The dataset includes over 80 features per flow that describe traffic time properties, packet directions, volume statistics, interarrival times, and protocol-level behaviors. Examples of features include Flow Duration, Total Fwd Packets, Flow Bytes/s, Fwd Packet Length Mean, Bwd IAT Std, Init Win Bytes Fwd, Active Mean, and Idle Std. Every record is labeled as either Benign or as one of several Attack Types (e.g., DDoS, Brute Force, Web Attack, etc.), allowing supervised machine learning techniques to be evaluated accurately.

Features of the dataset are given in Table A.3 as seen in the next page.

C. Examination of each feature in the Dataset

A. Flow Identification (Metadata):

Flow ID, Source IP, Source Port, Destination Port and Destination IP are unique identifiers. Protocol is the Protocol Number and Timestamp is the time the flow started.

B. Packet Counts & Size Statistics:

Total Fwd Packets / Total Bwd Packets are the count of packets in each direction.

Total Length of Bwd Packets / Total Length of Fwd Packets are the total size of the packets in both directions. Fwd/Bwd Packet Length (Max, Min, Mean, Std) are the statistical distributions of the packet sizes.

C. Inter Arrival Time (IAT):

Flow IAT (Mean, Max, Min, Std) are the times between packets in the flow.

Fwd IAT (Total, Min, Max, Std, Mean) are the times between packets in the forward direction.

Bwd IAT (Total, Min, Max, Std, Mean) are the times between packets in the backward direction.

D. Flag Counts:

Fwd/Bwd PSH Flags are Push Flag Counts.

Fwd/Bwd URG Flags are Urgent Flag Counts.

If we were to go in to more Specifics about each flag the table below would explain each flag clearly.

Flag	Meaning	High Count Often Indicates...
SYN	Start Connection	DoS (SYN Flood), Port Scan
RST	Abort Connection	Victim rejecting a Port Scan
FIN	End Connection	OS Fingerprinting, FIN Scan
PSH	Push Data Now	Interactive Botnet C&C, Slowloris DoS
URG	Urgent Data	"Xmas" Scans (highly suspicious)
ACK	Receipt Confirm	Normal traffic (if balanced), ACK Scan (if isolated)
ECE/CWE	Network Congestion	Volumetric DDoS (network saturation)

Table B.2 Adapted from [38]

E. Window & Header Size:

Fwd/Bwd Header Length are the sizes of the headers.

Init_Win_Bytes_Fwd / Init_Win_Bytes_Fwd are the TCP window sizes at the start of the flow.

F. Subflow & Active/Idle Stats:

Active (Mean, Max, Min, Std) are the times the flow was active until it became idle.

Idle (Mean, Max, Min, Std) are the times the flow was idle until it became active.

Subflow as can be understood from the meaning word is about features related to the number of packets from sub-segments of the flow.

II. LITERATURE SURVEY

A. Machine Learning Research on CIC-IDS2017

The CIC-IDS2017 dataset has become one of the most widely used benchmarks for machine-learning-based intrusion detection in recent years. Sharafaldin et al. (2016) introduced the dataset as a modern alternative to KDD'99-style benchmarks, with realistic benign traffic and a diverse set of attack scenarios including brute force, DoS/DDoS, web attacks, botnets, Heartbleed and infiltration. Subsequent exploratory studies have analysed its statistical properties, traffic composition and labelling strategy in order to understand how suitable it is for training data-driven intrusion detection systems.

A second recurring observation in the CIC-IDS2017 literature is the severe class imbalance between benign and attack flows. In the original dataset, benign traffic accounts for the vast majority of instances, while many attack categories appear only in very small numbers, which easily biases standard classifiers towards the majority class. Several works therefore apply data cleaning and re-labelling, followed by oversampling techniques such as random oversampling or SMOTE to rebalance the training data [X], [X]. Figure Z illustrates this process in the binary setting: before preprocessing, benign flows dominate the dataset; after cleaning, the imbalance remains; and after oversampling, the numbers of benign and attack instances are approximately equal. These results underline that explicit imbalance handling is a critical step before training machine-learning models on CIC-IDS2017.

TABLE A.1.: Representative supervised-learning baselines on CIC-IDS2017

Study (year)	Dataset(s)	Algorithms	Pre-processing	Best Reported Performance (overall)
Talukder et al. (2024)	CIC-IDS2017	DT, RF, ET	RO + PCA + stacking	≈99.99% accuracy
Maseer et al. (2021)	CIC-IDS2017	ANN, DT, k-NN NB, RF, SVM, CNN	Imbalanced multi-class setting	≈99% acc / F1
Injadat et al. (2021)	CIC-IDS2017	Tree-based ML Pipeline	Oversampling + feature selection + HPO	>99% accuracy

Table A.1. summarised from [5], [14]

TABLE A.2.: Binary frequency distribution of CIC-IDS2017 dataset

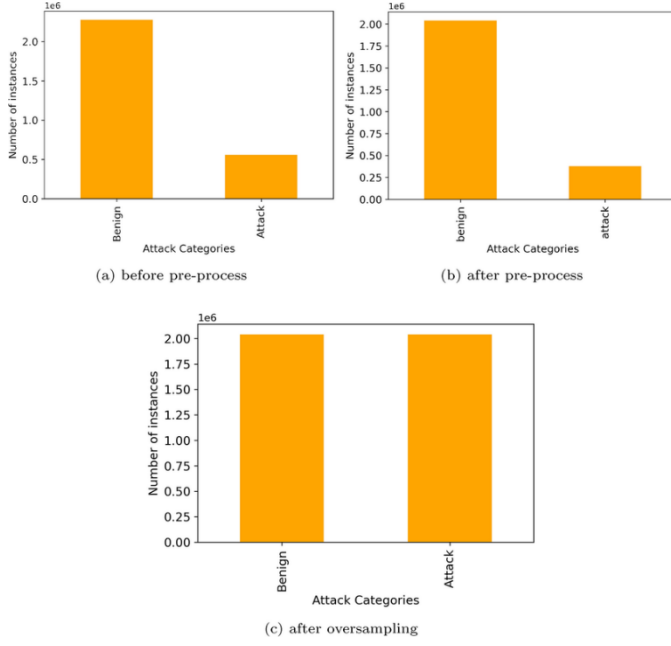


Table A.2. [5]

A second line of research investigates feature selection and dimensionality reduction to improve both performance and interpretability. Al-Daweri et al. used Information Gain to rank CIC-IDS2017 features and demonstrated that a relatively small subset of flows statistics (e.g. packet counts, inter-arrival times, byte and duration statistics) is sufficient for effective anomaly detection. Olanrewaju et al. and Kahraman & Çat further evaluated correlation-based selection and tree-based feature-importance scores, showing that models trained on 10–30 of the most informative features can match or even surpass the performance of models trained on the full feature set. These studies motivate treating feature engineering as a central design choice rather than simply feeding all available attributes into a classifier. They also report that using a reduced subset of features substantially lowers execution time, which confirms the computational benefits of dimensionality reduction on CIC-IDS2017 (Figure X).

Figure A.1. Execution time for different feature subsets on CIC-IDS2017,

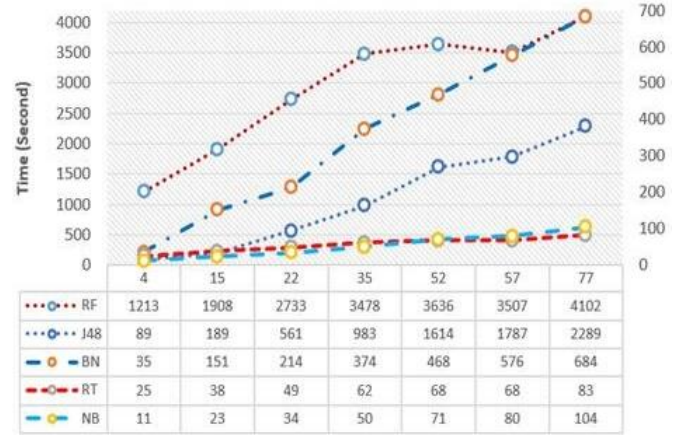


Figure A.1. adapted from Kurniabudi et al. [6].

One specific resource that is widely used as a reference for understanding the CIC-IDS2017 feature space is the summary table of flow attributes provided in [21]. This work lists all features generated by CICFlowMeter for the dataset and organises them according to their semantics, such as flow duration, packet- and byte-count statistics in forward and backward directions, inter-arrival time measures, header flag counters, and rate-based indicators. The table also clarifies the meaning and units of each attribute (for example, whether a value is per-flow, per-direction, or computed as an average), which helps to avoid misinterpretation during preprocessing and model design. By relying on [21] as a canonical description of the available features, researchers can more systematically select relevant attributes, document their feature-engineering choices, and ensure that different studies on CIC-IDS2017 are at least referring to the same underlying variables when they report results.

TABLE A.3. : Features in cic ids 2017 dataset

SNo	S No	Feature Name	SNo	Feature Name	SNo	Feature Name
1	22	Flow ID	43	Fwd Packets/s	64	Fwd Avg Packets/Bulk
2	23	Source IP	44	Bwd Packets/s	65	Fwd Avg Bulk Rate
3	24	Source Port	45	Min Packet Length	66	Bwd Avg Bytes/Bulk
4	25	Destination IP	46	Max Packet Length	67	Bwd Avg Packets/Bulk
5	26	Destination Port	47	Packet Length Mean	68	Bwd Avg Bulk Rate
6	27	Protocol	48	Packet Length Std	69	Subflow Fwd Packets
7	28	Timestamp	49	Packet Length Variance	70	Subflow Fwd Bytes
8	29	Flow Duration	50	FIN Flag Count	71	Subflow Bwd Packets
9	30	Total Fwd Packets	51	SYN Flag Count	72	Subflow Bwd Bytes
10	31	Total Backward Packets	52	RST Flag Count	73	Init. Win. bytes.forward
11	32	Total Length of Fwd Packets	53	PSH Flag Count	74	Init. Win. bytes.backward
12	33	Total Length of Bwd Packets	54	ACK Flag Count	75	act.data.pkt.fwd
13	34	Fwd Packet Length Max	55	URG Flag Count	76	min_seg_size.forward
14	35	Fwd Packet Length Min	56	CWE Flag Count	77	Active Mean
15	36	Fwd Packet Length Mean	57	ECN Flag Count	78	Active Std
16	37	Fwd Packet Length Std	58	Down/Up Ratio	79	Active Max
17	38	Bwd Packet Length Max	59	Average Packet Size	80	Active Min
18	39	Bwd Packet Length Min	60	Avg Fwd Segment Size	81	Idle Mean
19	40	Bwd Packet Length Mean	61	Avg Bwd Segment Size	82	Idle Std
20	41	Bwd Packet Length Std	62	Fwd Header Length	83	Idle Max
21	42	Flow Bytes/s	63	Fwd Avg Bytes/Bulk	84	Idle Min

Table A.3. [21]

Complementary to feature lists, Sari [22] conducts an exploratory data analysis of the CIC-IDS2017 dataset with the aim of assessing its suitability for intrusion detection research. The study examines the number of flows per capture day and per attack type, highlights the strong imbalance between benign and malicious traffic, and visualises the distribution of key features across classes. It also inspects correlations between attributes, identifies redundant or highly dependent

features, and points out the presence of missing, zero-only or extreme values that may bias learning algorithms if left untreated. Based on these findings, [22] recommends concrete preprocessing steps, such as cleaning corrupted rows, handling NaN and infinite values, normalising feature scales and carefully designing train–test splits that reflect the temporal structure of the captures. As a result, this work provides practical guidance for building more reliable machine-learning pipelines on CIC-IDS2017 and supports the argument that the dataset should be treated as raw measurement data requiring careful preparation rather than as a ready-to-use benchmark.

TABLE A.4.: Dataset feature space and sample size & Summary of Suggested Solutions for tackling the Issues in the Dataset

Capture	Name Chosen for The Data Set Capture	No Input Features	No Samples/Instances
Capture 1	FriAfternoonPortScan	78	286,467
Capture 2	FriAfternoonDDOS	78	225,745
Capture 3	FriMorning	78	191,033
Capture 4	MonMorningHour	78	529,918
Capture 5	ThurAfternoonInfiltration	78	288,602
Capture 6	ThursdayMornWebAttacks	78	170,366
Capture 7	TueWorking	78	445,909
Capture 8	WedHour	78	692,703

Dataset Name	Brief Information about the Dataset	Class Distribution	Data Types	Attack Types	Available formats	Unknown or Missing Values?	Open Problems Observed in the Data Set
Captures CICIDS2017 data set	The data set is usually used as a reference for intrusion detection studies.	Imbalance class distribution	Numeric (a mixture of integer and floating-point data types)	Several types of attacks of different magnitudes are captured in the data set.	Netflow and PCAPs. The format used for all of the experimental analyses in this paper is net flow.	YES, there are few unknown or missing values in each group of the dataset. That is, each of the sets of datasets has NaN values that have to be addressed before being used to build models.	The data set is very large and contains complex data patterns and high-class imbalance, the input variable values are on different scaling.

Table A.4. [22]

More recent work has shifted towards deep learning and hybrid architectures. Ferrag et al. surveyed deep learning approaches for intrusion detection and reported that convolutional and recurrent networks can exploit complex non-linear relationships in the CIC-IDS2017 features, especially when combined with careful preprocessing and class-imbalance handling. Gu and Lu proposed an SVM classifier with a learned feature embedding, where a neural network is used to transform flow features into a more separable latent space before classification. Antunes et al. benchmarked several deep architectures, including CNNs and LSTMs, against classical machine-learning models and found that deep models are particularly beneficial for multi-class scenarios and for attacks with subtle behavioural patterns. Ali et al. introduced an improved CNN-based intrusion detection model and reported strong results on CIC-IDS2017 when appropriate normalization and tuning are applied.

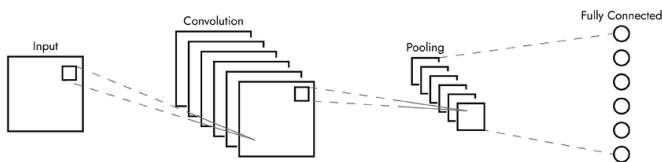


Fig. A.2. Convolutional neural network architecture [11], [13]

In parallel, application-specific and evaluation-focused studies have examined CIC-IDS2017 under more constrained or realistic conditions. Farhat et al. concentrated on DoS/DDoS detection and compared different machine-learning techniques

specifically on the corresponding attack subsets, highlighting the impact of class imbalance and metric choice on the perceived performance. Verma & Ranga explored the use of CIC-IDS2017 in IoT-oriented intrusion detection settings, whereas technical reports such as the Tinexta Defence study considered NetFlow-style representations and operational deployment aspects. Open-source implementations and example pipelines, for instance the GitHub repository by Pervez, have further contributed to standardising preprocessing and evaluation choices used by many subsequent researchers.

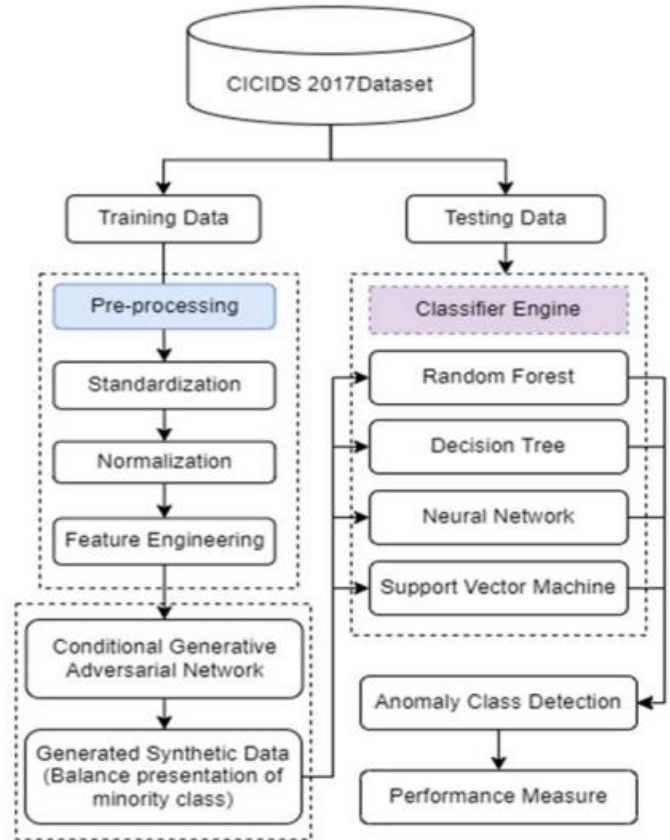


Figure A.3.

Overall, the existing work on CIC-IDS2017 converges towards a fairly consistent end-to-end machine-learning pipeline, illustrated in Figure A.3. After data collection, the raw flows are cleaned, merged and reduced, followed by feature scaling and categorical encoding; the severe class imbalance is then handled through re-sampling techniques such as random oversampling or SMOTE. Many successful studies further apply feature selection or embedding and, in some cases, dimensionality reduction (e.g. PCA) before training tree-based ensembles or neural network models with k-fold cross-validation. Although individual design choices differ, this general structure is common across the most competitive approaches and effectively turns CIC-IDS2017 into a de facto benchmark for machine-learning-based intrusion detection. These studies collectively establish CIC-IDS2017 as a de facto benchmark for machine-learning-based intrusion detection and

provide the empirical foundation for the key insights and best practices stated in the following subsection.

B. Key Insights and Best Practices from the Literature

Recent work on CIC-IDS2017 shows that achieving strong intrusion-detection performance is less about selecting the “fanciest” model and more about treating the dataset carefully, engineering high-quality features, and evaluating models in a security-aware way. Across traditional machine learning and deep learning studies, several common strategies consistently lead to reliable and reproducible results.

1. Treat CIC-IDS2017 as imperfect raw data, not a ready-made benchmark

CIC-IDS2017 was designed to update older IDS benchmarks with realistic traffic, multiple operating systems and a wide range of attack scenarios (brute force, DoS/DDoS, web attacks, botnets, Heartbleed, infiltration). However, detailed analyses reveal a number of artefacts introduced by the capture setup and by CICFlowMeter. Mislabeled flows, flows created from TCP termination (“TCP appendices”), protocol-detection errors and coarse time-window-based labelling can all inflate performance if they are not corrected.

Label	Original	Intermediate (appendices fixed)	Final (payload filter)
BENIGN	2271326	1823964	1823964
ATTEMPTED*	0	0	447362
FTP-Patator	7934	3984	3973
SSH-Patator	5898	2988	2980
DoS GoldenEye	10293	7647	7567
DoS Hulk	230124	159048	158469
DoS Slowhttptest	5499	5109	1742
DoS slowloris	5791	5707	4001
Heartbleed	11	11	11
Web Brute Force	1507	1365	151
Web Attack XSS	652	679	27
Web Attack SQL	21	12	12
Infiltration	36	48	32
Bot	1956	2208	738
PortScan	158842	159023	159023
DDoS	128022	95123	95123

TABLE B.1. Flow counts in the original, intermediate, and final cleaned versions of CIC-IDS2017. Adapted from Engelen et al., 2021 [23]

Best practice: before any modelling, researchers carefully inspect how the traffic was generated, check the relationship between PCAPs and flow records, and either relabel or remove problematic flows (for example empty connections, obviously corrupted records or purely “attempted” attacks). Some studies even build on improved derivatives such as LYCOS-IDS2017 to reduce these artefacts.

2. Robust data cleaning and preprocessing

A second, very consistent theme is that disciplined preprocessing is essential. Multiple papers report rows containing only zeros, NaN or Infinity values, often caused by divisions by zero in rate-based features when the flow duration

is zero. Others identify features that are constant or clearly miscomputed (e.g. some backward flag and bulk-rate fields).

Label	Original			Intermediate			Final		
	Pr	Re	F1	Pr	Re	F1	Pr	Re	F1
BENIGN	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
FTP-Patator	0.99	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00
SSH-Patator	0.99	0.98	0.99	1.00	0.99	0.99	1.00	0.99	0.99
DoS GoldenEye	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99
DoS Hulk	0.99	0.99	0.99	0.99	0.99	0.99	1.00	0.99	0.99
DoS Slowhttptest	0.98	0.99	0.98	0.97	0.99	0.98	0.99	0.99	0.99
DoS slowloris	0.99	0.99	0.99	0.99	0.99	0.99	1.00	0.99	0.99
Heartbleed*	1.00	0.67	0.77	1.00	0.67	0.78	1.00	0.89	0.93
Web Brute Force	0.70	0.85	0.77	0.69	0.91	0.78	1.00	0.91	0.95
Web Attack XSS*	0.52	0.11	0.18	0.52	0.16	0.25	1.00	0.67	0.79
Web Attack SQL*	0.33	0.07	0.11	0.00	0.00	0.00	0.00	0.00	0.00
Infiltration*	0.96	0.70	0.81	1.00	0.50	0.66	1.00	0.83	0.91
Bot	0.92	0.45	0.60	0.79	0.40	0.53	1.00	0.99	0.99
PortScan	0.99	0.99	0.99	0.96	0.97	0.97	0.96	0.97	0.97
DDoS	0.99	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00
Accuracy	0.99			0.99			0.99		
Weighted Avg	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99	0.99

Table B.2. Flow features extracted by CICFlowMeter for the CIC-IDS2017 dataset. Adapted from ResearchGate Dataset Table, 2021 [23]

Effective pipelines therefore:

- remove fully empty or clearly corrupted rows;
- handle NaN/ ∞ values via explicit imputation (e.g. replacing them with a meaningful statistic such as the column maximum or mean) instead of blindly dropping rows;
- drop constant or obviously erroneous attributes;
- exclude identifiers such as Flow ID, IP addresses, ports and sometimes timestamps when the goal is a generalizable IDS, so that models learn behavioural patterns rather than memorizing specific hosts.

Because feature scales vary dramatically (for example, flow duration vs. packet counts), normalization is standard. Min-max scaling to $[0,1]$ or z-score normalization is used to stabilize training, especially for neural networks and SVMs, and to avoid a few large-magnitude features dominating the learning process.

3. Systematic handling of severe class imbalance

CIC-IDS2017 is highly imbalanced: benign traffic and a few attack types dominate, while classes such as Heartbleed and Infiltration appear only rarely.

Category	Label	Sample	Feature
Benign	Benign	2,273,097	
DoS/DDoS	DoS Hulk, DDoS,	380,699	Dest Port, Flow Duration, Tot Fwd/Bwd Pkts, Tot Len Fwd/Bwd Pkts, Fwd/Bwd Pkt Len Max(Min(Mean(Std, Min)Max Pkt Len, Pkt Len Mean(Std) Var, Avg Pkt Size, Avg Fwd/Bwd Seg Size, Flow Bytes/Pkts, Fwd/Bwd Pkts, Fwd/Bwd Avg Bulk Rate, Fwd/Bwd Avg Bytes/Pkts, Flow IAT Mean) SlpMax(Min, Fwd/Bwd IAT TotLen(Mean(Std)Max(Min, Down-Up Ratio, ActiveIdle Mean(Std)Max(Min, Fwd/Bwd PSH Flags, Fwd/Bwd URG Flags, Init Win Bytes Fwd/Bwd, Fwd/Bwd Header Len, Act Data Pkt Fwd, FIN(SYN/RST)IP(HACK)URG)CWE/ECE Flags, Min Seg Size Fwd, Subflow Fwd/Bwd Pkts/Bytes, Label)
	DoS GoldenEye,		
	DoS Slowloris,		
	DoS Slowhttptest, Heartbleed		
PortScan	PortScan	158,930	
Brute Force	FTP-Patator, SSH-Patator	13,835	
Web Attack	Web Attack-Brute Force,	2180	
	Web Attack-XSS,		
	Web Attack-SQL Injection		
Botnet	Bot	1966	
Total	-	2,850,797	79

TABLE B.3. Class distribution of the CIC-IDS2017 dataset showing the imbalance across attack categories. Adapted from Electronics (MDPI), 2024 [40]

If this skew is ignored, models can obtain very high overall accuracy while failing almost completely on minority attacks.

To address this, successful studies combine several strategies:

- Stratified sampling and splitting: train/validation/test sets are built so that all attack types are represented, rather than splitting each CSV file independently by day.
- Oversampling of minority classes: SMOTE or related synthetic oversampling techniques are widely adopted to boost rare attacks (Infiltration, Web attacks, Heartbleed) in the training data, while keeping the test set untouched.
- Class-weighted loss functions: for tree ensembles, SVMs and neural networks, setting class weights to “balanced” or using focal / gradient-guided losses improves recall on rare classes without excessively overfitting to the majority.

These practices shift the optimisation target from “99% accuracy” to high recall and F1-score for minority attacks, which is more aligned with operational IDS requirements.

4. Feature selection and dimensionality reduction

With more than 80 flow features, naive use of the full feature set increases computation and often leads to overfitting. The literature consistently shows that a compact subset of well-chosen features is sufficient for high performance.

Common approaches include:

- filter methods such as Information Gain, correlation analysis or χ^2 statistics to rank features;
- wrapper or embedded methods using tree-based models (Random Forest, Extra Trees, XGBoost) to compute feature importance and iteratively remove low-contribution attributes;

No.	Weight	Feat. ID	Feat. Names	No.	Weight	Feat. ID	Feat. Names
1	0.7521	41	Packet Length Std	40	0.337	29	Bwd IAT Mean
2	0.7197	13	Total Length of Bwd Packets	41	0.3242	7	Bwd Packets/s
3	0.7197	65	Subflow Bwd Bytes	42	0.3237	19	Bwd Packet Length Min
4	0.6937	66	Init_Win_bytes_forward	43	0.2874	69	min_seg_size_forward
5	0.6916	63	Subflow Fwd Bytes	44	0.2843	76	Idle Max
6	0.6916	12	Total Length of Fwd Packets	45	0.2781	74	Idle Mean
7	0.6823	42	Packet Length Variance	46	0.2773	27	Fwd IAT Min
8	0.6694	40	Packet Length Mean	47	0.2757	77	Idle Min
9	0.6571	18	Bwd Packet Length Max	48	0.2752	70	Active Mean
10	0.6511	39	Max Packet Length	49	0.2717	31	Bwd IAT Min
11	0.6472	67	Init_Win_bytes_backward	50	0.2711	72	Active Max
12	0.6401	52	Average Packet Size	51	0.2705	73	Active Min
13	0.64	20	Bwd Packet Length Mean	52	0.2596	38	Min Packet Length
14	0.64	54	Avg Bwd Segment Size	53	0.2573	15	Fwd Packet Length Min
15	0.6313	14	Fwd Packet Length Max	54	0.2489	68	act_data_pkt_fwd
16	0.6096	8	Destination Port	55	0.2463	23	Flow IAT
17	0.6089	22	Flow IAT Max	56	0.2174	6	Bwd IAT Std
18	0.5835	9	Flow Duration	57	0.1319	46	PSH Flag Count
19	0.5769	55	Fwd Header Length	58	0.0955	51	Down/Up Ratio
20	0.5707	26	Fwd IAT Max	59	0.0743	47	ACK Flag Count
21	0.5485	36	Bwd Header Length	60	0.0545	75	Idle Std

Table B.4 Feature importance ranking for CIC-IDS2017 based on Information Gain scores. Adapted from IJEEI Feature Selection Study, 2021 [26]

- dimensionality-reduction techniques such as PCA or autoencoders, particularly in deep-learning pipelines, to compress features while preserving most of the variance.

Empirical results show that between 10 and 30 carefully selected features (for example, packet length statistics, inter-arrival times, flow duration, forward packet counts, initial window bytes) are sufficient to reach or approach state-of-the-art detection performance, while significantly reducing training time and improving interpretability.

5. Modelling strategies: strong baselines and specialised deep models

The reviewed work covers both traditional machine-learning models and deep architectures. Despite the popularity of deep learning, several robust conclusions emerge:

- Tree-based ensembles (Random Forest, Extra Trees, Gradient Boosting, XGBoost) repeatedly achieve excellent accuracy, precision and recall on CIC-IDS2017 flow features. They are often competitive with or superior to more complex networks, while being faster to train and easier to tune.
- Classical models such as Logistic Regression, SVM, Decision Trees, k-NN and Naïve Bayes remain important baselines. When combined with good preprocessing, cross-validation and feature selection, they provide strong performance and clear decision boundaries, which is attractive for deployment.

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)	FPR (%)	MCC
LDA	88.10	81.63	98.33	89.21	22.12	0.7785
QDA	97.72	96.00	99.59	97.76	4.15	0.9550
SVM	96.98	95.78	98.30	97.02	4.33	0.9400
k-NN	99.31	98.87	98.75	98.61	1.10	0.9864
DT	99.88	99.87	99.89	99.88	0.13	0.9976
RF	99.90	99.90	99.91	99.90	0.10	0.9981
MLP	99.10	98.44	99.79	99.11	1.58	0.9821

TABLE B.5. PERFORMANCE METRICS OF CLASSICAL MACHINE-LEARNING MODELS ON CIC-IDS2017. ADAPTED FROM SCITEPRESS INTRUSION DETECTION STUDY (2022) [24]

Deep learning strategies bring additional benefits in specific scenarios. Feed-forward MLPs trained on cleaned, balanced and reduced feature sets can match or exceed tree ensembles. CNN- and LSTM-based models that operate either on transformed flow vectors (e.g. reshaping 1-D feature vectors into 2-D matrices) or on packet sequences can capture spatial and temporal dependencies that hand-crafted features miss, which is useful for detecting stealthy or low-rate attacks. Hybrid CNN–LSTM architectures are particularly effective for modelling both content and temporal structure.

Several studies also show that attack-specific configurations can be beneficial. For example, converting the problem to binary “benign vs. DoS/DDoS” and using a small set of high-weight features allows gradient-boosting models such as XGBoost to reach very high accuracy with extremely low false-alarm rates and reasonable training times—an attractive trade-off for real-time deployments.

Overall, the literature suggests a pragmatic strategy: start with interpretable, tree-based or MLP baselines on engineered flow features; then consider deep, packet- or sequence-based architectures when cross-dataset generalization or detection of subtle temporal patterns becomes a priority.

Reference	ML Technique	F1 Score	FS Method	N Attr.	Comments
Sharafaldin et al. [25]	ID3	0.980	RandomForest Regressor	4 per class	Weighted evaluation metrics
Engelen et al. [48]	RF	0.990	NA *	NA *	
Rosay et al. [49]	RF	0.999	Not used	74	Dest. port included. Regeneration and labeling of flows
Abdulhammed et al. [50]	RF	0.988	PCA	10	IP addresses and ports included
Kurniabudi et al. [51]	RF	0.998	Information Gain ranker	15	Dest. port included
Meemongkolkiat et al. [52]	Bagging ensemble	0.999	Recursive Feature Elimination & RF	18	Dest. and source ports included. 30 percent of the dataset
	PART/148	0.999	Not used	77	
Our work	RF/148	0.990	CFS	6	
	PART/148	0.997	Not used	14	Zeeko-based flows and attributes

* NA = Not available.

Table B.6. Performance results of related works in the CICIDS2017 dataset. Adapted from Meemongkolkiat et al., Sensors (2022), and related studies. [41]

6. Evaluation practices aligned with security objectives

Because of the dataset’s imbalance and artefacts, evaluation design has a large impact on reported results. Successful work adopts several common practices:

- reporting class-aware metrics such as precision, recall, F1-score and false-positive rate (FPR), rather than relying solely on overall accuracy;
- using confusion matrices and per-class metrics to analyse which attacks remain difficult to detect;
- employing k-fold cross-validation or clearly separated train/validation/test splits to reduce overfitting;
- distinguishing between attempted and successful attacks in labelling when possible, so that the IDS is evaluated on meaningful malicious flows rather than on failed connection attempts.

In addition, temporal characteristics of CIC-IDS2017 (different days and scenarios) are considered during evaluation. For example, some studies use the Monday benign-only file to learn normal behaviour and then test on mixed attack days, while others merge daily files and perform stratified sampling to obtain representative and memory-efficient subsets.

7. Emerging directions: explainability and spatio-temporal analysis

Finally, recent research highlights two important trends:

- Explainable AI (XAI): Techniques such as SHAP and LIME are integrated into IDS pipelines to explain why a given flow is classified as malicious. This improves analyst trust, helps debug models that accidentally learn artefacts (for example, specific IPs), and accelerates incident investigation.
- Spatio-temporal deep models: Hybrid CNN–LSTM architectures are used to jointly model spatial relations between features and temporal evolution of traffic, improving detection of complex multi-stage or low-rate attacks that are difficult to capture with purely static flow-based models.

III. CONCLUSION

Taken together, the literature indicates that high-quality cybersecurity analysis on CIC-IDS2017 is achieved when researchers:

1. critically inspect and repair the dataset rather than treating it as perfect;
2. apply systematic preprocessing, including imputation and normalization;
3. explicitly correct class imbalance and label quality issues;
4. reduce the feature space to the most informative network-behaviour indicators;
5. combine strong, interpretable baselines with carefully tuned deep models where appropriate; and
6. evaluate models with metrics and experimental setups that reflect real operational constraints.

These best practices shift the focus from maximising benchmark accuracy to building trustworthy, data-aware and operationally meaningful intrusion-detection models, which is the central goal of cybersecurity-oriented data mining on CIC-IDS2017.

REFERENCES

- [1] I. Sharafaldin, A. H. Lashkari, and A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization," in *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*, Portugal, Jan. 2018, pp. 108–116.
- [2] Canadian Institute for Cybersecurity, "Intrusion detection evaluation dataset (CIC-IDS2017)," University of New Brunswick, 2017.
- [3] B. A. Tama, "Overview and Exploratory Analyses of CICIDS 2017 Intrusion Detection Dataset," *Data*, vol. 8, no. 1, 2023. doi: 10.3390/data8010000.
- [4] G. Engelen, O. Rimmer, and W. Joosen, "Troubleshooting an Intrusion Detection Dataset: the CICIDS2017 Case Study," in *2021 IEEE Security and Privacy Workshops (SPW)*, San Francisco, CA, USA, 2021, pp. 7–12.
- [5] M. A. Talukder, et al., "Machine Learning-based Intrusion Detection for CICIDS2017," in *Proceedings of the 10th International Conference on Model-Driven Engineering and Software Development (MODELSWARD)*, 2022, pp. 1–8.
- [6] K. M. A. Al-Daweri, et al., "CICIDS-2017 Dataset Feature Analysis With Information Gain for Anomaly Detection," *IEEE Access*, vol. 8, pp. 1–15, 2020.
- [7] R. F. Olanrewaju, B. U. I. Khan, and M. H. Habaebi, "Important Features of CICIDS-2017 Dataset for Network Intrusion Detection," *Indonesian Journal of Electrical Engineering and Informatics (IJEI)*, vol. 10, no. 3, pp. 1–9, 2022.
- [8] M. A. Ferrag, et al., "Deep Learning for Cyber Security Intrusion Detection: Approaches, Datasets, and Comparative Study," *Mathematics*, vol. 10, no. 24, p. 4745, 2022.
- [9] J. Gu and S. Lu, "An Effective Intrusion Detection Approach Using SVM with Naïve Bayes Feature Embedding," *Sensors*, vol. 22, no. 23, p. 9326, 2022.
- [10] H. T. Kahraman and B. Çat, "Feature Analysis of CICIDS2017 for Intrusion Detection," *Turkish Journal of Electrical Engineering and Computer Sciences*, vol. 30, no. 1, pp. 1–15, 2022.
- [11] M. Antunes, L. Oliveira, and A. S. Gomes, "Benchmarking Deep Learning Methods for Behaviour-Based Network Intrusion Detection," *Informatics*, vol. 9, no. 1, p. 29, 2022.
- [12] S. Farhat, M. Abdelkader, A. Meddeb-Makhlouf, and F. Zarai, "Evaluation of DoS/DDoS Attack Detection with ML Techniques on CIC-IDS2017 Dataset," in *Proceedings of the 9th International Conference on Information Systems Security and Privacy (ICISSP)*, Lisbon, Portugal, 2023, pp. 287–295.
- [13] M. H. Ali, et al., "An Improving Intrusion Detection Model Based on Novel CNN Technique Using CIC-IDS2017 Dataset," in *2023 IEEE International Conference on Communication Networks and Satellite (Comnetsat)*, 2023, pp. 1–6.
- [14] A. Verma and V. Ranga, "Benchmarking of Machine Learning Algorithms for Intrusion Detection Systems in IoT," *IEEE Access*, vol. 8, pp. 1–12, 2020.
- [15] N. Pervez, "Intrusion-Detection-CICIDS2017," GitHub Repository, 2021.
- [16] Tinexta Defence, "NetFlowMeter Analysis of the CICIDS2017 Dataset and Intrusion Detection via ML," Technical Report, 2022.
- [17] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, "Kitsune: An Ensemble of Autoencoders for Online Network Intrusion Detection," in *Network and Distributed System Security Symposium (NDSS)*, 2018.
- [18] S. El Hag and A. Fernandez, "Optimized Machine Learning for Intrusion Detection," *Electronics*, vol. 13, no. 15, p. 3014, 2024.
- [19] Z. Chiba, et al., "Implementation of Machine Learning Algorithms on CICIDS-2017 Dataset for Intrusion Detection Using WEKA," in *International Conference on Networking and Advanced Systems (ICNAS)*, 2019.
- [20] IEEE Dataport, "CICIDS2017," Available: <https://iee-dataport.org/documents/cicids2017>.
- [21] ResearchGate, "Features in CIC-IDS2017 dataset,".
- [22] A. Sari, "Overview and Exploratory Analyses of CICIDS2017 Intrusion Detection Dataset," ResearchGate preprint, 2023.
- [23] WTMIC Workshop, "Troubleshooting the CICIDS2017 Dataset," 2021.
- [24] Scitepress, "Intrusion Detection Using CICIDS2017," 2022.
- [25] SciSpace, "CICIDS-2017 Dataset Feature Analysis with Information Gain".
- [26] Pratama *et al.*, "Important Features of CICIDS2017 Dataset," University of Sriwijaya Repository, 2021.
- [27] TÜBİTAK Journal of Electrical Engineering, "Intrusion Detection Techniques," Available: <https://journals.tubitak.gov.tr/cgi/viewcontent.cgi?article=3802&context=elektrik>.
- [28] Dergipark, "Intrusion Detection Evaluation Studies".
- [29] Dergipark, "Flow-based Intrusion Detection Approaches," Available: <https://dergipark.org.tr/en/download/article-file/4694162>.

- [30] Polbeng Repository, “ML-based Intrusion Detection Analysis,” 2020.
- [31] TEM Journal, “CICIDS2017 Detection Techniques,” 2023.
- [32] INRIA HAL Archive, “Flow Reconstruction in IDS Datasets,” 2021.
- [33] ResearchSquare, “Deep Learning for Intrusion Detection,” 2023.
- [34] ACM Digital Library, “Performance of ML Algorithms on CICIDS2017,” 2023.
- [35] Springer, “Hybrid Deep Learning for Intrusion Detection,” 2024.
- [36] arXiv, “Deep Learning Approaches to Intrusion Detection,” 2024.
- [37] arXiv, “Advanced CICIDS2017 Attack Detection,” 2025.
- [38] Postel, J. (1981). Transmission Control Protocol. IETF RFC 793.
- [39] Iman Sharafaldin, Arash Habibi Lashkari, and Ali A. Ghorbani, “Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization”, 4th International Conference on Information Systems Security and Privacy (ICISSP), Portugal, January 2018
- [40] M. Deng, C. Sun, Y. Kan, H. Xu, X. Zhou, and S. Fan, “Network intrusion detection based on deep belief network broad equalization learning system,” *Electronics*, vol. 13, no. 15, p. 3014, 2024.
- [41] M. Rodríguez, Á. Alesanco, L. Mehavilla, and J. García, “Evaluation of machine learning techniques for traffic flow-based intrusion detection,” *Sensors*, vol. 22, no. 23, p. 9326, 2022.